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第 27 章 Quantum Statistics

§27.1 Three-dimensional indefinite well

System Hamiltonian,

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + U(x, y, z), \quad (27.1.1)$$

where

$$U(x, y, z) = \begin{cases} 0, & 0 < x < a, 0 < y < b, 0 < z < c \\ +\infty, & \text{Otherwise.} \end{cases} \quad (27.1.2)$$

$$\hat{H}\psi(x, y, z) = E\psi(x, y, z). \quad (27.1.3)$$

$$\psi(x, y, z) = \begin{cases} \text{nonzero,} & 0 < x < a, 0 < y < b, 0 < z < c \\ 0, & \text{Otherwise.} \end{cases} \quad (27.1.4)$$

Within potential well,

$$-\frac{\hbar^2}{2m}\nabla^2\psi(x, y, z) = E\psi(x, y, z), \quad (27.1.5)$$

where

$$\psi(0, y, z) = 0, \quad \psi(a, y, z) = 0, \quad (27.1.6)$$

$$\psi(x, 0, z) = 0, \quad \psi(x, b, z) = 0, \quad (27.1.7)$$

$$\psi(x, y, 0) = 0, \quad \psi(x, y, c) = 0. \quad (27.1.8)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \quad (27.1.9)$$

$$-\frac{\hbar^2}{2m} \left[\frac{\partial^2}{\partial x^2}\psi(x, y, z) + \frac{\partial^2}{\partial y^2}\psi(x, y, z) + \frac{\partial^2}{\partial z^2}\psi(x, y, z) \right] = E\psi(x, y, z). \quad (27.1.10)$$

$$\frac{\partial^2}{\partial x^2}\psi(x, y, z) + \frac{\partial^2}{\partial y^2}\psi(x, y, z) + \frac{\partial^2}{\partial z^2}\psi(x, y, z) = -k^2\psi(x, y, z), \quad (27.1.11)$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}}. \quad (27.1.12)$$

$$\psi(x, y, z) = \psi(x)\psi(y)\psi(z), \quad (27.1.13)$$

$$\psi(y)\psi(z)\frac{\partial^2\psi(x)}{\partial x^2} + \psi(x)\psi(z)\frac{\partial^2\psi(y)}{\partial y^2} + \psi(x)\psi(y)\frac{\partial^2\psi(z)}{\partial z^2} = -k^2\psi(x)\psi(y)\psi(z), \quad (27.1.14)$$

$$\frac{1}{\psi(x)}\frac{\partial^2\psi(x)}{\partial x^2} + \frac{1}{\psi(y)}\frac{\partial^2\psi(y)}{\partial y^2} + \frac{1}{\psi(z)}\frac{\partial^2\psi(z)}{\partial z^2} = -k^2, \quad (27.1.15)$$

$$\frac{1}{\psi(x)}\frac{\partial^2\psi(x)}{\partial x^2} = -k^2 - \frac{1}{\psi(y)}\frac{\partial^2\psi(y)}{\partial y^2} - \frac{1}{\psi(z)}\frac{\partial^2\psi(z)}{\partial z^2} \equiv -k_1^2, \quad (27.1.16)$$

$$\frac{1}{\psi(x)}\frac{\partial^2\psi(x)}{\partial x^2} = -k_1^2, \quad (27.1.17)$$

$$-k^2 - \frac{1}{\psi(y)}\frac{\partial^2\psi(y)}{\partial y^2} - \frac{1}{\psi(z)}\frac{\partial^2\psi(z)}{\partial z^2} = -k_1^2, \quad (27.1.18)$$

$$\frac{\partial^2\psi(x)}{\partial x^2} + k_1^2\psi(x) = 0, \quad (27.1.19)$$

$$\frac{1}{\psi(y)}\frac{\partial^2\psi(y)}{\partial y^2} = -k^2 + k_1^2 - \frac{1}{\psi(z)}\frac{\partial^2\psi(z)}{\partial z^2} \equiv -k_2^2, \quad (27.1.20)$$

$$\frac{\partial^2\psi(x)}{\partial x^2} + k_1^2\psi(x) = 0, \quad (27.1.21)$$

$$\frac{1}{\psi(y)}\frac{\partial^2\psi(y)}{\partial y^2} = -k_2^2, \quad (27.1.22)$$

$$-k^2 + k_1^2 - \frac{1}{\psi(z)}\frac{\partial^2\psi(z)}{\partial z^2} = -k_2^2, \quad (27.1.23)$$

$$\frac{\partial^2\psi(x)}{\partial x^2} + k_1^2\psi(x) = 0, \quad (27.1.24)$$

$$\frac{\partial^2\psi(y)}{\partial y^2} + k_2^2\psi(y) = 0, \quad (27.1.25)$$

$$\frac{\partial^2\psi(z)}{\partial z^2} + k_3^2\psi(z) = 0, \quad (27.1.26)$$

where

$$k_3^2 = k^2 - k_1^2 - k_2^2. \quad (27.1.27)$$

$$\frac{\partial^2 \psi(x)}{\partial x^2} + k_1^2 \psi(x) = 0, \quad (27.1.28)$$

$$\frac{\partial^2 \psi(y)}{\partial y^2} + k_2^2 \psi(y) = 0, \quad (27.1.29)$$

$$\frac{\partial^2 \psi(z)}{\partial z^2} + k_3^2 \psi(z) = 0, \quad (27.1.30)$$

where

$$k^2 = k_1^2 + k_2^2 + k_3^2, \quad E = \frac{\hbar^2}{2m} k^2. \quad (27.1.31)$$

$$\psi(x, y, z) = \psi(x)\psi(y)\psi(z), \quad (27.1.32)$$

$$\psi(0, y, z) = 0, \quad \psi(a, y, z) = 0 \Rightarrow \psi(x)|_{x=0}\psi(y)\psi(z) = 0, \quad \psi(x)|_{x=a}\psi(y)\psi(z) = 0, \quad (27.1.33)$$

$$\psi(x, 0, z) = 0, \quad \psi(x, b, z) = 0 \Rightarrow \psi(x)\psi(y)|_{y=0}\psi(z) = 0, \quad \psi(x)\psi(y)|_{y=b}\psi(z) = 0, \quad (27.1.34)$$

$$\psi(x, y, 0) = 0, \quad \psi(x, y, c) = 0 \Rightarrow \psi(x)\psi(y)\psi(z)|_{z=0} = 0, \quad \psi(x)\psi(y)\psi(z)|_{z=c} = 0. \quad (27.1.35)$$

$$\psi(x)|_{x=0} = 0, \quad \psi(x)|_{x=a} = 0, \quad (27.1.36)$$

$$\psi(y)|_{y=0} = 0, \quad \psi(y)|_{y=b} = 0, \quad (27.1.37)$$

$$\psi(z)|_{z=0} = 0, \quad \psi(z)|_{z=c} = 0. \quad (27.1.38)$$

$$\frac{\partial^2 \psi(x)}{\partial x^2} + k_1^2 \psi(x) = 0, \quad (27.1.39)$$

$$\psi(x)|_{x=0} = 0, \quad \psi(x)|_{x=a} = 0, \quad (27.1.40)$$

$$\frac{\partial^2 \psi(y)}{\partial y^2} + k_2^2 \psi(y) = 0, \quad (27.1.41)$$

$$\psi(y)|_{y=0} = 0, \quad \psi(y)|_{y=b} = 0, \quad (27.1.42)$$

$$\frac{\partial^2 \psi(z)}{\partial z^2} + k_3^2 \psi(z) = 0, \quad (27.1.43)$$

$$\psi(z)|_{z=0} = 0, \quad \psi(z)|_{z=c} = 0. \quad (27.1.44)$$

$$\psi(x) = \sqrt{\frac{2}{a}} \sin(k_1^{(n)} x), \quad (27.1.45)$$

$$k_1^{(n)} = n \frac{\pi}{a}, \quad 0 < n \in \mathbb{Z}, \quad (27.1.46)$$

$$\psi(y) = \sqrt{\frac{2}{b}} \sin(k_2^{(m)} y), \quad (27.1.47)$$

$$k_2^{(m)} = m \frac{\pi}{b}, \quad 0 < m \in \mathbb{Z}, \quad (27.1.48)$$

$$\psi(z) = \sqrt{\frac{2}{c}} \sin(k_3^{(l)} z), \quad (27.1.49)$$

$$k_3^{(l)} = l \frac{\pi}{c}, \quad 0 < l \in \mathbb{Z}. \quad (27.1.50)$$

$$\psi(x, y, z) = \psi(x)\psi(y)\psi(z). \quad (27.1.51)$$

$$\psi_{nml}(x, y, z) = \sqrt{\frac{8}{abc}} \sin(k_1^{(n)} x) \sin(k_2^{(m)} y) \sin(k_3^{(l)} z). \quad (27.1.52)$$

$$k^2 = k_1^2 + k_2^2 + k_3^2, \quad E = \frac{\hbar^2}{2m} k^2. \quad (27.1.53)$$

$$\mathbf{k} = k_1 \mathbf{e}_x + k_2 \mathbf{e}_y + k_3 \mathbf{e}_z, \quad E = \frac{\hbar^2}{2m} \mathbf{k}^2. \quad (27.1.54)$$

$$k_{nml} = \sqrt{\left(n \frac{\pi}{a}\right)^2 + \left(m \frac{\pi}{b}\right)^2 + \left(l \frac{\pi}{c}\right)^2}, \quad E = \frac{\hbar^2}{2m} k_{nml}^2. \quad (27.1.55)$$

$$k_1^{(n)} = n \frac{\pi}{a}, \quad 0 < n \in \mathbb{Z} \Rightarrow \Delta k_1 = \frac{\pi}{a}, \quad (27.1.56)$$

$$k_2^{(m)} = m \frac{\pi}{b}, \quad 0 < m \in \mathbb{Z} \Rightarrow \Delta k_2 = \frac{\pi}{b}, \quad (27.1.57)$$

$$k_3^{(l)} = l \frac{\pi}{c}, \quad 0 < l \in \mathbb{Z} \Rightarrow \Delta k_3 = \frac{\pi}{c}. \quad (27.1.58)$$

$$\Delta \mathbf{k} = \Delta k_1 \Delta k_2 \Delta k_3 = \frac{\pi^3}{abc} = \frac{\pi^3}{V}. \quad (27.1.59)$$

$$\begin{aligned} \frac{1}{V} \sum_{\mathbf{k}} f(\mathbf{k}) &= \frac{1}{V} \frac{1}{\Delta \mathbf{k}} \sum_{\mathbf{k}} f(\mathbf{k}) \Delta \mathbf{k} \\ &= \frac{1}{\pi^3} \sum_{\mathbf{k}} f(\mathbf{k}) \Delta \mathbf{k}. \end{aligned} \quad (27.1.60)$$

$$\begin{aligned}
\lim_{V \rightarrow +\infty} \frac{1}{V} \sum_{\mathbf{k}} f(\mathbf{k}) &= \lim_{V \rightarrow +\infty} \sum_{\mathbf{k}} f(\mathbf{k}) \Delta \mathbf{k} \\
&= \frac{1}{\pi^3} \int f(\mathbf{k}) d\mathbf{k} \\
&= \frac{1}{\pi^3} \int_0^{+\infty} dk_1 \int_0^{+\infty} dk_2 \int_0^{+\infty} dk_3 f(\mathbf{k}). \tag{27.1.61}
\end{aligned}$$

$$f(\mathbf{k}) = f(k), \quad e.g., \quad f(\mathbf{k}) = f(\varepsilon_{\mathbf{k}}) = f(\varepsilon_k) = f\left(\frac{\hbar^2}{2m} k^2\right). \tag{27.1.62}$$

$$\begin{aligned}
\lim_{V \rightarrow +\infty} \frac{1}{V} \sum_{\mathbf{k}} f(\mathbf{k}) &= \frac{1}{\pi^3} \int_0^{+\infty} dk_1 \int_0^{+\infty} dk_2 \int_0^{+\infty} dk_3 f(k) \\
&= \frac{1}{\pi^3} \frac{1}{8} \int_{-\infty}^{+\infty} dk_1 \int_{-\infty}^{+\infty} dk_2 \int_{-\infty}^{+\infty} dk_3 f(k) \\
&= \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} dk_1 \int_{-\infty}^{+\infty} dk_2 \int_{-\infty}^{+\infty} dk_3 f(k) \\
&= \int \frac{d\mathbf{k}}{(2\pi)^3} f(\mathbf{k}). \tag{27.1.63}
\end{aligned}$$

$$\lim_{V \rightarrow +\infty} \frac{1}{V} \sum_{\mathbf{k}} f(\mathbf{k}) = \int \frac{d\mathbf{k}}{(2\pi)^3} f(\mathbf{k}). \tag{27.1.64}$$

$$\begin{aligned}
\lim_{V \rightarrow +\infty} \frac{1}{V} \sum_{\mathbf{k}} f(\mathbf{k}) &= \int \frac{d\mathbf{k}}{(2\pi)^3} f(\mathbf{k}) \\
&= \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} dk_1 \int_{-\infty}^{+\infty} dk_2 \int_{-\infty}^{+\infty} dk_3 f(k) \\
&= \frac{1}{(2\pi)^3} \int_0^{+\infty} dk \int_0^\pi d\theta \int_0^{2\pi} d\varphi k^2 \sin \theta f(k) \\
&= \frac{4\pi}{(2\pi)^3} \int_0^{+\infty} dk k^2 f(k). \tag{27.1.65}
\end{aligned}$$

$$f(\mathbf{k}) = f(\varepsilon_{\mathbf{k}}) = f(\varepsilon_k) = f\left(\frac{\hbar^2}{2m} k^2\right). \tag{27.1.66}$$

$$\begin{aligned}
\lim_{V \rightarrow +\infty} \frac{1}{V} \sum_{\mathbf{k}} f(\mathbf{k}) &= \frac{4\pi}{(2\pi)^3} \int_0^{+\infty} dk k^2 f(\varepsilon_k) \\
&= \frac{4\pi}{(2\pi)^3} \frac{1}{2} \int_0^{+\infty} dk^2 k f(\varepsilon_k) \\
&= \frac{2\pi}{(2\pi)^3} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^{+\infty} d\varepsilon \varepsilon^{1/2} f(\varepsilon) \\
&= \frac{1}{(2\pi)^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^{+\infty} d\varepsilon \varepsilon^{1/2} f(\varepsilon). \tag{27.1.67}
\end{aligned}$$

$$\lim_{V \rightarrow +\infty} \frac{1}{V} \sum_{\mathbf{k}} f(\mathbf{k}) = \int_0^{+\infty} d\varepsilon D(\varepsilon) \varepsilon^{1/2} f(\varepsilon), \quad (27.1.68)$$

where

$$D(\varepsilon) = \frac{1}{(2\pi)^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \varepsilon^{1/2}. \quad (27.1.69)$$

$$D(\varepsilon) = \frac{2\pi}{h^3} (2m)^{3/2} \varepsilon^{1/2}. \quad (27.1.70)$$

§27.2 Quantum distribution

Fermi-Dirac distribution:

$$f(\varepsilon) = \frac{1}{e^{\beta(\varepsilon-\mu)} + 1}, \quad \beta = \frac{1}{k_B T}. \quad (27.2.1)$$

Bose-Einstein distribution:

$$f(\varepsilon) = \frac{1}{e^{\beta(\varepsilon-\mu)} - 1}, \quad \beta = \frac{1}{k_B T}. \quad (27.2.2)$$

When $\exp(-\beta\mu) \gg 1$,

$$f(\varepsilon) = e^{-\beta(\varepsilon-\mu)}. \quad (27.2.3)$$

$$\begin{aligned} n &= \frac{N}{V} \\ &= \frac{1}{V} \sum_{\mathbf{k}} f(\varepsilon_{\mathbf{k}}) \\ &= \frac{1}{V} \sum_{\mathbf{k}} e^{-\beta(\varepsilon_{\mathbf{k}}-\mu)} \\ &= e^{\beta\mu} \frac{1}{V} \sum_{\mathbf{k}} e^{-\beta\varepsilon_{\mathbf{k}}}. \end{aligned} \quad (27.2.4)$$

$$\begin{aligned} n &= e^{\beta\mu} \lim_{V \rightarrow +\infty} \frac{1}{V} \sum_{\mathbf{k}} e^{-\beta\varepsilon_{\mathbf{k}}} \\ &= e^{\beta\mu} \int_0^{+\infty} d\varepsilon D(\varepsilon) e^{-\beta\varepsilon} \\ &= e^{\beta\mu} \frac{2\pi}{h^3} (2m)^{3/2} \int_0^{+\infty} d\varepsilon \varepsilon^{1/2} e^{-\beta\varepsilon} \\ &= e^{\beta\mu} \frac{2\pi}{h^3} (2mk_B T)^{3/2} \int_0^{+\infty} dx x^{1/2} e^{-x} \\ &= e^{\beta\mu} \frac{2\pi}{h^3} (2mk_B T)^{3/2} \Gamma(3/2) \\ &= e^{\beta\mu} \frac{2\pi}{h^3} (2mk_B T)^{3/2} \frac{\sqrt{\pi}}{2} \\ &= e^{\beta\mu} \frac{1}{h^3} (2\pi mk_B T)^{3/2} \\ &= e^{\beta\mu} \lambda^{-3/2}, \end{aligned} \quad (27.2.5)$$

where

$$\lambda = \frac{h}{\sqrt{2\pi m k_B T}}. \quad (27.2.6)$$

$$e^{\beta\mu} = n\lambda^{3/2}. \quad (27.2.7)$$

$$e^{\beta\mu} \ll 1, \quad (27.2.8)$$

$$n\lambda^{3/2} \ll 1. \quad (27.2.9)$$

λ : thermal wavelength.